

## 9 2026-02-02 | Week 04 | Lecture 09

### 9.1 Conditional probability

*This section is based on section 2.4 in the text.*

**Definition 29** (Conditional probability). Let  $A, B$  be events, and assume that  $\mathbb{P}[A] > 0$ . Then the **conditional probability of  $B$ , given  $A$** , denoted  $\mathbb{P}[B | A]$ , is given by the formula

$$\mathbb{P}[B | A] = \frac{\mathbb{P}[A \cap B]}{\mathbb{P}[A]}.$$

**Interpretation:**  $\mathbb{P}[A | B]$  is the probability of  $A$  when we know that event  $B$  happened.

**Multiplication rule:**  $\mathbb{P}[A \cap B] = \mathbb{P}[B | A] \mathbb{P}[A]$

**Definition 30.** We say that there exists a **positive relationship** between events  $A$  and  $B$  if

$$\mathbb{P}[B | A] > \mathbb{P}[B],$$

and a **negative relationship** if

$$\mathbb{P}[B | A] < \mathbb{P}[B].$$

- Smoking and getting lung cancer have a positive relationship: if you know someone is a smoker ( $A$ ), then the probability that they get lung cancer ( $B$ ) is higher than if you didn't know they were a smoker.

**Example 31** (Conditional probability). I roll a dice behind a screen. I tell you that I rolled an even number. What's the probability that the dice roll is a 4 or a 6?

**Solution:** We have  $S = \{1, 2, 3, 4, 5, 6\}$ ,  $A = \{2, 4, 6\}$ ,  $B = \{4, 6\}$ . We want to compute  $\mathbb{P}[B | A]$ . By Definition 29,

$$\mathbb{P}[B | A] = \frac{\mathbb{P}[AB]}{\mathbb{P}[A]} = \frac{2/6}{1/2} = \frac{2}{3}.$$

Put differently, we have three outcomes in  $A$ , which are all equally likely. Two of them (4 and 6) mean that event  $B$  occurs. So by the enumeration principle, the probability is  $2/3$ .

End of Example 31.  $\square$

**Example 32** (Similar to examples 2.27 and 2.28 in the textbook). A vampire goes to the blood bank looking to find some type O+ blood (the most delicious type). He finds 4 unlabeled bags of blood. Only one of the bags is O+, but the vampire doesn't know which one, so he resorts to taste-testing to find the O+ bag.

- (a) What is the probability that he must test at least 3 bags to find the desired type?

**Solution:** Let  $X$  be the number of tested bags. We want to find  $\mathbb{P}[X \geq 3]$ . Let  $A = [\text{first bag isn't O+}]$ . Let  $B = [\text{second bag isn't O+}]$

$$\begin{aligned}\mathbb{P}[X \geq 3] &= \mathbb{P}[A \cap B] \\ &= \mathbb{P}[A] \mathbb{P}[B | A] \\ &= \frac{3}{4} \cdot \frac{2}{3} \\ &= \frac{1}{2}.\end{aligned}$$

- (b) What's the probability that he must test exactly 3 bags?

**Solution:** We want  $\mathbb{P}[X = 3] = \mathbb{P}[\text{first bag isn't O+} \cap \text{second bag isn't O+} \cap \text{third bag is O+}]$ .

$$\begin{aligned}
\mathbb{P}[X = 3] &= \mathbb{P}[\text{third is } \mid \text{first isn't} \cap \text{second isn't}] \times \mathbb{P}[\text{second isn't} \mid \text{first isn't}] \cdot \mathbb{P}[\text{first isn't}] \\
&= \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \\
&= \frac{1}{4}
\end{aligned}$$

Here we have used the following multiplication rule:

$$\begin{aligned}
\mathbb{P}[A \cap B \cap C] &= \mathbb{P}[C \mid A \cap B] \mathbb{P}[A \cap B] \\
&= \mathbb{P}[C \mid A \cap B] \mathbb{P}[B \mid A] \mathbb{P}[A]
\end{aligned}$$

with  $A = [\text{first bag isn't O+}]$ ,  $B = [\text{second bag isn't O+}]$ , and  $C = [\text{third bag is O+}]$ .

End of Example 32.  $\square$